WAHYU FHALDIAN

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Semarang, Central Java, Indonesia

Informatics Engineering graduate specializing in Intelligent Systems, with interests in Data Science, Machine Learning, and Artificial Intelligence. Experienced in data processing, Machine Learning, Computer Vision, and Explainable AI through academic and project-based work.

EDUCATION

Universitas Dian Nuswantoro (UDINUS) **Semarang, Indonesia**
Bachelor of Computer Science | GPA: 3.57/4.00 **Sep 2022 - May 2026**

- Machine Learning
- Computer Vision
- Natural Language Processing
- Artificial Intelligence
- Data Processing
- Intelligent Systems

EXPERIENCE

Diskominfo Kota Semarang **Semarang, Indonesia**
Backend Developer: SPBE - Intern **Oct - Nov2025**

- Contributed to the development of a web-based platform to support esports community management
- Collaborated with the frontend team as a Backend Developer to develop and integrate backend services
- Developed REST APIs and GET/POST endpoints for data exchange and management
- Managed team, committee, and community information using MongoDB
- Initial development focused on supporting the Udinus esports community within the scope of the program

SELECTED PROJECTS

Gizi Play **Dec 2025**
Interactive Web-Based Nutrition Education

- Technologies used HTML, CSS, JavaScript
- Developed an interactive web application for nutrition education based on client requirements
- Implemented interface logic and responsive behavior using native JavaScript
- Delivered the application on schedule for use in field-based educational activities

Poverty Prediction Using Explainable Machine Learning **Oct 2025**
Sinkron Journal - Published Research

- Developed Machine Learning models to predict poverty levels across 35 regencies and cities in Central Java using socioeconomic data from Statistics Indonesia (BPS).
- Performed data preprocessing, training and testing, model development, and evaluation using MAE, RMSE, and R^2.
- Compared six Machine Learning approaches XGBoost, Random Forest, CatBoost, LightGBM, Elastic Net Regression, and Stacking Ensemble.
- Applied Explainable AI (XAI) techniques using SHAP, PDP, and ALE to interpret model predictions and identify influential factors affecting poverty levels.
- DOI: 10.33395/sinkron.v9i4.15312

Bruang Play **Sep 2025**
Augmented Reality Application

- Technologies used Unity, Vuforia Engine, Android
- Developed an Android-based Augmented Reality application using Unity and Vuforia Engine
- Integrated 3D models to provide interactive visualizations within the application

Tomato Disease Classifier **Jun 2025**

- Technologies used Python, TensorFlow/Keras, CNN, Streamlit

- Developed an image classification application for detecting tomato leaf diseases using DenseNet201, VGG16, and ResNet50
- Built a Streamlit interface for image upload, 224×224 preprocessing, class prediction, and prediction probability display

Water Quality Prediction **Dec 2024**
- Technologies used Python, Random Forest, Pandas, Scikit-learn, Streamlit
- Developed a Streamlit application to predict whether water is suitable for consumption based on parameters including pH, hardness, solids, sulfate, conductivity, and turbidity
- Integrated a trained Random Forest model and scaler for input preprocessing, prediction, and probability display

CERTIFICATIONS

Associate Data Scientist **Apr 2026**
Dian Nuswantoro University Professional Certification Institution: LSP

Center of Foreign Language Training **Jan 2026**
Dian Nuswantoro University

AI Literacy for Everyone: Understand, Apply, Create **Oct 2025**
BASS Learning and Development Institute: LS BASS

GettingStarted with Deep Learning **Jan 2025**
NVIDIA Deep Learning Institute (DLI)

TECHNICAL SKILLS

Programming
Python, JavaScript, HTML, CSS

Machine Learning & Data Science
Scikit-learn, TensorFlow/Keras, Pandas, Random Forest, XGBoost, CatBoost, PCA, SMOTE, SHAP

Backend & Database
REST API, GET/POST API, MongoDB, MongoDB Atlas

AI & Computer Vision
Machine Learning, Computer Vision, Natural Language Processing, Image Classification, Deep Learning, Augmented Reality

Tools
Git, GitHub, Streamlit, Unity, Vuforia Engine

LANGUAGES

Indonesian: Native
English: Basic